

Regression and Residuals

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Error Analysis

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Accuracy and Imbalance

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Train a Classifier

Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Explain cancellation for Regression and residuals. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Regression and residuals. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Observed [6, 10, 13], predicted [8, 9, 11] cm. Give residuals, mean residual and error directions.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Actual leaf length is 7 cm and prediction is 10 cm. Using $\text{residual} = \text{actual} - \text{predicted}$, calculate residual and absolute error. Interpret the sign.

6. Solve the given case

Actual height=12 cm, prediction=9 cm. Residual means actual minus predicted.

Calculate and interpret.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Define residual as actual minus predicted. Actual height is 14 cm and prediction is 17 cm. Calculate residual and interpret its sign.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 19 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Use a miniature growth-prediction side experiment to distinguish regression evidence from the main classifier metrics.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Actual leaf length is 7 cm and prediction is 10 cm. Using $\text{residual} = \text{actual} - \text{predicted}$, calculate residual and absolute error. Interpret the sign.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Calculate residuals

Compute the worked residuals and identify overprediction.

My reasoning and evidence

13. W2 Explain cancellation

Why does mean residual zero not imply perfect predictions?

My reasoning and evidence

14. W3 Read a pattern

Interpret residuals -2, 0, 3 ordered by growth stage.

My reasoning and evidence

15. W4 Apply independently

Observed [6, 10, 13], predicted [8, 9, 11] cm. Give residuals, mean residual and error directions.

My reasoning and evidence

16. W5 Check units

A report combines 2 cm and .02 m as if they were different magnitudes. Repair it.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Regression and Residuals

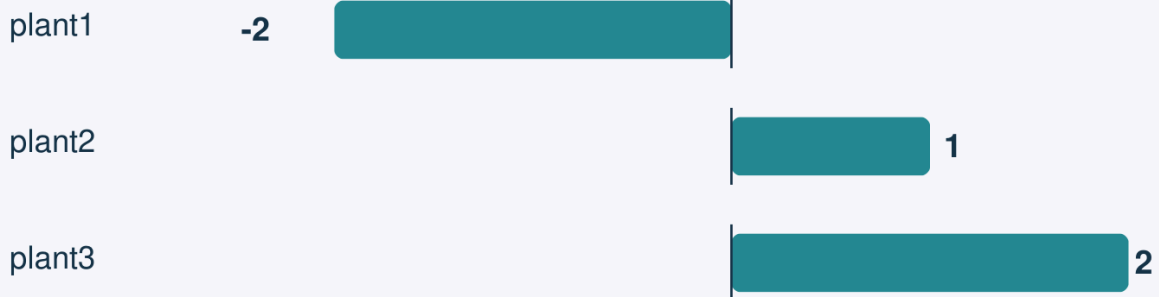
Use residual signs to say whether each plant-height estimate is too high or low.

Residual=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Observed minus predicted height

cm



Bar length shows the supplied value; the marked baseline is zero.

Regression and Residuals

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Subtract matched pairs

Residuals are $5-7=-2$, $8-7=1$, $12-10=2$ cm.



2. Read directions

First prediction is too high by 2; second too low by 1; third too low by 2.



3. Avoid cancellation

Mean residual is $1/3$ cm; MAE is $5/3$ cm, so small signed mean does not imply small errors.

CHANGE ONE THING

Raise all predictions by 1 cm.

Explain the changed result using the matching lesson.

Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Use residual signs to say whether each plant-height estimate is too high or low.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Mean residual is $\frac{1}{3}$ cm; MAE is $\frac{5}{3}$ cm, so small signed mean does not imply small errors.

1. Read directions
2. Avoid cancellation
3. Subtract matched pairs

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Avoid cancellation Mean residual is $\frac{1}{3}$ cm; MAE is $\frac{5}{3}$ cm, so small signed mean does not imply small errors. Read the picture in order. Mean residual is $\frac{1}{3}$ cm; MAE is $\frac{5}{3}$ cm, so small signed mean does not imply small errors. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Raise all predictions by 1 cm. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Residuals become $[-3, 0, 1]$; mean = $-\frac{2}{3}$ cm and MAE = $\frac{4}{3}$ cm.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Residuals become $[-3, 0, 1]$; $\text{mean} = -2/3$ cm and $\text{MAE} = 4/3$ cm. Each residual decreases by 1 under the observed-minus-predicted convention.

Repair a missing building block

Review Error Analysis (ai-4/18). Goal: Categorize development errors using evidence. Short practice: What evidence would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Train a Classifier (ai-4/13). Goal: Connect training-only scaling and neighbour prediction in one pipeline. Short practice: Which pipeline record would help another team reproduce the result? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
